NaviMaster: Learning a Unified Policy for GUI and Embodied Navigation Tasks

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Paper version: arXiv 2508.02046v4 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Recent advances in GUI and embodied navigation have driven progress, yet these domains have largely evolved in isolation, with disparate datasets and training paradigms*

Two stacks, two action spaces, no cross-task transfer. The paper casts both tasks as **one MDP** over egocentric views and pixel-level targets.

Why unify

1. **Two models.** Separate agents double cost.
2. **Weak OOD transfer.** Task-specific scaling fails OOD.
3. **Sparse rewards.** Binary RFT zeroes most rollouts.

Evaluation setup

10 GUI + 4 afford. sets · ObjectNav (Habitat) · Qwen2.5VL-7B base · GPT-4o thought labels · 20k steps (10k+10k) · GRPO · 3 epochs · 8×A800

- GUI results are **offline** per-step SR + grounding; ObjectNav in **simulation**.
- OOD sets disjoint from training; Odyssey in-domain.

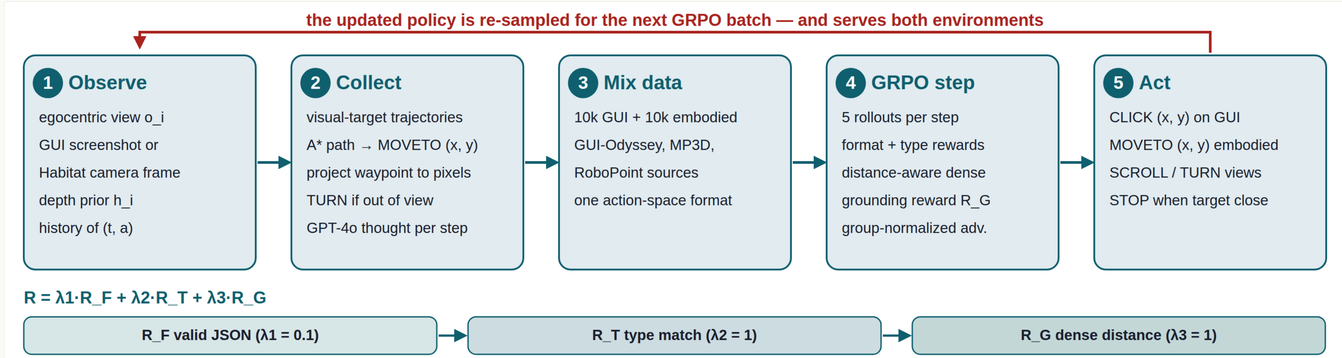
Our read (CUA Radar)

63	★★	online
score of 100	band	evidence level

- **Reports:** OOD held-out sets; SFT + RL baselines; ablations; code + data.
- **Flags:** variance unclear; online eval partial; data-amount match only.
- Pattern R01: single-step GRPO, rule rewards. Authors' numbers.

Method: one policy, visual-target trajectories, dense-reward GRPO

- **Sees / acts:** egocentric view + depth + (thought, action) history; unified space: BACK/HOME/STOP, SCROLL/TURN, CLICK(x,y) / MOVETO(x,y).
- **Data:** GUI-Odyssey reuse; Habitat A* paths projected to pixel targets; GPT-4o writes ~60-word first-person thoughts.
- **Training:** R1-Zero GRPO on single steps of the 20k mix (10k GUI + 10k embodied); EasyR1, 8×A800, 3 epochs.



$$R_G = \left(1 - \frac{d_j}{\theta_d}\right) \left[\, d_j < \theta_d,\ p_j < \theta_h \,\right]$$

Dense grounding reward (weight λ3 = 1): pixel distance d_j, gated by depth disparity p_j

GUI step success rate (%) — offline

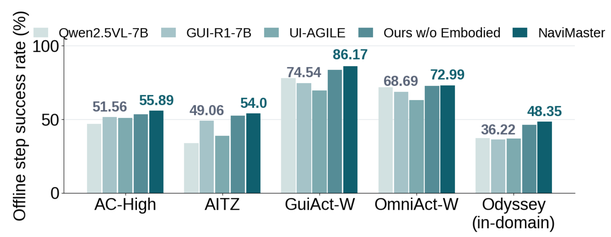


Table 1, redrawn: offline per-step SR. Odyssey in-domain; others OOD.

AC-High SR 55.89 vs 51.56 (GUI-R1-7B)

SO WHAT → *GR: 93.90 AC-Low, 78.15 AC-High.*

Ablation: ObjectNav success rate (%)

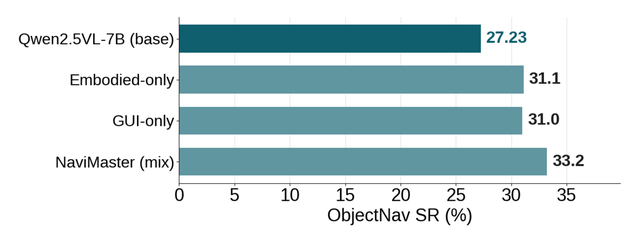


Table 3, redrawn: vary the training data source; Habitat simulation, unseen validation branch.

SO WHAT → *Mix beats both single-source policies, and both beat the base (SPL 12.60 vs 9.68). Same order on affordance: 77.34 vs 67.86 (GUI-only).*

Headline Numbers

77.34%
RoboRefIt affordance SR
vs 49.82 for RoboPoint-13B on the same offline benchmark

- **33.20%** ObjectNav SR (sim)
- **55.89** AC-High step SR
- **12.60** ObjectNav SPL

The paper's own controls

CONTROL	NAVIMASTER	COMPARED WITH
Dense vs sparse (AC-High)	55.89	sparse: 54.07
Mix vs single (ObjectNav)	33.20	single: 31.00 / 31.10
3B base (AC-High)	45.26	single: 28.75
Threshold θd (AC-High)	200 px: 55.89	100–400: 53.61–55.77

Takeaway

One GRPO policy over mixed GUI + embodied visual-target steps; the reported gains are offline step metrics plus simulated navigation, not live-device tests.

7B · 20k samples · R1-Zero: no SFT stage · dense > sparse reward · GPT-4o labels thoughts · code + data released

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